The Last Teacher Dies

After teacher removal, learners solved 73.3% of frozen food-latch attempts on two unseen maps. The same survivors solved 52.2% with their inherited weights and 50.6% when messages were blocked.

Survival and performance on unseen maps

Each frozen result shows correct attempts divided by resolved attempts across three seeds and two maps. Population counts pool the three seeds. Attempts from the same seed or individual are dependent.

Source arm	Foster lineage alive	Original fosters / 36	Resident lineage alive	Acquired weights	Inherited weights	Channel blocked
Intact, residents removed	89	26	0	384/524 (73.3%)	269/515 (52.2%)	270/534 (50.6%)
Disabled	99	26	0	279/532 (52.4%)	281/547 (51.4%)	291/587 (49.6%)
Shuffled	102	24	0	244/517 (47.2%)	244/519 (47.0%)	273/518 (52.7%)
Teacher-isolated	101	25	0	333/546 (61.0%)	270/516 (52.3%)	261/518 (50.4%)
Randomized teachers	95	25	0	296/484 (61.2%)	276/519 (53.2%)	257/514 (50.0%)
Intact, residents retained	26	21	91	295/424 (69.6%)	203/409 (49.6%)	228/435 (52.4%)

What persisted after removal

The intact removal arm beat teacher isolation on each seed. Acquired-weight accuracy was 73.0%, 79.5%, and 65.0%, compared with 53.7%, 67.0%, and 62.8% under teacher isolation. Randomized teachers did better on the third seed. Isolation still allowed peer and individual learning.

Residents delivered 132 packets to unrelated fosters during contact, with 75 successful latches. After removal, fosters delivered 482 messages to other fosters. Of those, 313 crossed ancestry roots and 254 crossed generation numbers. Among the 26 surviving original fosters, 112 of 156 preferred sender or receiver choices stayed the same from cutoff to endpoint.

What happened when teachers stayed

Forty-five of 48 original residents survived in the retained arm. After the cutoff, residents delivered 95 messages to fosters, and fosters delivered 99 to residents. The arm ended with 91 resident-lineage and 26 foster-lineage organisms, compared with 89 foster-lineage organisms after removal. Retention also increased competition for food and population slots.

The pilot supports transmission of useful signals across the removal boundary. Continued peer learning may also contribute. The pilot did not establish stable culture across generations or a population survival benefit.

How the experiment worked

Matched cohorts in a continuous world

Each run started with 16 freely moving residents. At tick 250, twelve newborns from four unrelated donor families entered. At tick 460, resident lineages were removed in five arms and retained in a sixth. The terrain, food stocks, learners, and clock continued through both boundaries. All 18 runs reached tick 660.

The six arms were intact, disabled, shuffled, teacher-isolated, randomized teachers, and intact with residents retained. Genotypes and sibling families matched within each seed. After the run, surviving original fosters were evaluated on two fixed unseen maps with learning and reproduction disabled. Each panel used matched bodies and starting positions across acquired, inherited, and blocked variants.

Individual signaling and inheritance

Four symbols started with no preset meanings. A scout saw a local food-latch marker that the receiver could not see. Sends cost 0.035 energy, traveled up to three Manhattan tiles, and took two ticks. Feedback followed the latch outcome. Nearby observers could learn from public successes, with a cap of 24 observation updates per individual.

Movement used the trained 32-module brain with its lifetime updates frozen during the pilot. Communication weights could change during life. Newborns inherited communication genes and initialized their own private weights. The intact removal arm recorded 119 natural births and reached generation 4. In 102 births the inherited communication genome differed from the parent.

CPU and verification

Standalone benchmark	Ticks	CPU seconds	Ticks / CPU second	Brain energy
32 modules	240	2.30	104.5	14.43
67 modules	240	4.69	51.2	25.36

The pilot used 279.3 process CPU seconds and about 28.1 million modeled neural operations. The 108 frozen evaluations used another 64.1 CPU seconds. All 22 regression tests passed. Data checks covered matching cohorts, ancestry, channel masks, packet delays, and energy balance.

Stage II

Replace removed residents with matched naive residents at equal density. Test learners immediately before and after removal with frozen weights, then follow specific conventions in descendants. That would separate teacher experience from crowding and continued peer learning.

Evidence files are in the Genesis 1.0.1 pilot directory under results.csv, transmission.csv, heldout.csv, and information.csv.